LEXSI LABS AI SAFETY & ALIGNMENT HACKATHON · TRACK 2b: REPAIRING SAFETY DRIFT WITHOUT RETRAINING

Same model, different language, different safety

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The problem

Tiny Aya's regional models average almost the same safety as Global, so a check on averages finds nothing. The tech report gives one clue: each model's **worst** language on MultiJail is 87.0 for Global, 78.1 for Fire and 77.5 for Earth (Table 7), and it doesn't name that language. We set out to find it. Our first pilot looked in the models' own regions and found no gap, because we were scoring refusals. Scored the report's way, by answer content, the gap appears in a language neither model was tuned for.

What the pilot found (planning signal, not a result)

Javanese, safe answers	Global	Fire	Earth	Water	10-language mean (G / F / E / W)
With Tiny Aya's preamble	87.0%	80.9%	77.1%	92.4%	94.9 / 94.1 / 91.6 / 94.9
Without the preamble	91.6%	80.9% (p=0.003)	**74.8%** (p<0.001)	89.3%	95.0 / 94.6 / 93.5 / 94.2

131 prompts per language, repair split only, 4-bit weights, two local 8B judges standing in for Command A. Without the preamble Earth's gap survives correction for all 60 comparisons, and Fire's only for 10 languages. Water shows no Javanese gap, which fits a gap that follows a model's region, though placing Javanese in Water's region is our reading of its card. Water dips in Bengali instead (85.5% and 84.7%, not significant). **Caution:** both judges call only 3 or 4 of Earth's 131 Javanese answers unsafe. The rest is judge disputes and breakdown: a blind read of 153 answers found many are repetition loops (19.8% of Earth's, 7.6% of Global's). The true gap is unknown until held-out data.

What we are doing

Step	What happens	Lexsi kit
1. Data	Harmful prompts (Aya red-teaming, MultiJail) and harmless look-alikes (XSTest) in English, Arabic, Swahili, Javanese, Bengali, Thai and Korean. Splits are by item, so a translation of a test prompt can never be used for the repair.	CuratorKIT
2. Confirm	At full precision on held-out prompts, per language: content safety, wrongful refusal of harmless prompts, and general ability. No repair claim unless the gap passes a corrected test.	AuditKIT
3. Locate	EAP-IG finds the safety circuit per language and per model. Where did regional tuning change it?	CircuitKIT
4. Repair	Five edits compared head to head, with settings tuned on the repair split only.	SafeTune
5. Verify	Re-test all three axes, then again after 4-bit compression (NF4 and GGUF Q4_K_M).	AuditKIT

The repair comparison

Arm	Edit	Why it is there
ReSta / Merge-back	Earth + α·(Global − Base), or a blend toward Global, whole model	Weight-space baselines
Circuit arms	**The same two vectors, only in the top 20% of layers by CircuitKIT score**	**Our main bet. Paired with its whole-model twin, so the difference isolates what the map adds**
Random layers	The best circuit edit on random layers outside the circuit, 3 seeds	The control: does the map beat editing fewer layers?
Steering	Refusal direction added at run time	Known to cause over-refusal; we check on Tiny Aya

How we will judge success (fixed before any repair)

- The gap must first be confirmed: full precision, held-out prompts, each preamble condition, paired CI above zero, corrected p below 0.05.
- Then close at least **half** of it, with wrongful refusal up by **at most 2 points** in any language and ability down by **at most 1 point**, all still holding at 4-bit.
- Every arm is reported, including losing ones and negative results.

Where we are today

The pipeline has run end to end on real Earth and Fire models at a small sample, and those runs fixed several crashes and silent errors. Next: confirm the gap on held-out data, with Fire and Water alongside as context, then run the repairs. We have also prepared six fixes for the kits (none submitted yet).

What we would value your view on

- Is content safety, as in the report, the right primary measure? And should a model that breaks down into repetition count as a safety failure, as the report's rubric implies, or be reported only as a separate failure?
- Do you know which language is behind the report's 78.1, and are our 8B judges a reasonable stand-in for Command A?
- Is a public multilingual eval set (prompts and labels only) acceptable under Cohere's acceptable-use policy?